make
based on credit
thousands)

get

approved

nt it was a
del.

)

w3)

w3

Tutorial-02

An Email service provider wants to calculate classify incomming emails as spam or not spam using a single-layer network model with 5 Hidden neuron.

x_1	x_2	x_3	Target(y)
5	3	1	1
1	0	1	0
7	5	1	1
2	1	1	0
6	4	1	1

Step-1 Initialize weights and bias

$w_1 = 0.1$; $w_2 = 0.1$; $w_3 = 0.1$; $b = 0$

Activation function = Sigmoid $= \frac{1}{1+e^{-z}}$

Step-2 $(x_1, x_2, x_3) \rightarrow (5, 3, 1)$

hidden layer = 5 neurons each with same weights

output = 1 neuron (spam).

= 0 (not spam)

Step-3 Formula for hidden neurons each hidden neuron computes

$$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + w_n x_n + b$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Step-4 5 hidden neurons all give the same value of h

$$\text{output}(s) = 0.1(h+h+h+h+h) = 0.5h$$

$$y_p = \sigma(s)$$

Step-5

z = 0.1(x1+x2+x3)	h sigmoid(z)	s (0.5)h	$y_p = \sigma(s)$
1) 0.1(5+3+1) 0.1(9) = 0.9	0.719	0.355	0.588
2) 0.1(2) = 0.2	0.5498	0.275	0.568
3) 0.1(7+5+1) 0.1(13) = 1.3	0.7858	0.393	0.597
4) 0.1(2+1+1) 0.1(4) = 0.4	0.5987	0.295	0.574
5) 0.1(6+4+1) 0.1(11) = 1.1	0.7503	0.375	0.592

$$\text{Error} = \frac{1}{2}(y - y_p)^2$$

$$E_1 = \frac{1}{2}(1-0.568)^2 = \frac{1}{2}(0.412)^2 = 0.084872$$

$$E_2 = \frac{1}{2}(0-0.568)^2 = \frac{1}{2}(-0.568)^2 = 0.161312$$

$$E_3 = \frac{1}{2}(1-0.597)^2 = \frac{1}{2}(0.403)^2 = 0.08120$$

$$E_4 = \frac{1}{2}(0-0.574)^2 = \frac{1}{2}(-0.574)^2 = 0.164738$$

$$E_5 = \frac{1}{2}(1-0.592)^2 = \frac{1}{2}(0.408)^2 = 0.0832$$

sample	y	y_p	Error
1	1	0.568	0.084
2	0	0.568	0.161312
3	1	0.597	0.081
4	0	0.574	0.1647
5	1	0.592	0.0832

→ If neuron output is near 1 than email is spam

→ If neuron output is near 0 than email is not-spam.

2) A bank uses a perception to decide whether loan should be approved. Build Network 5 neurons in hidden layer.

x_1	x_2	x_3	y
6	5	1	1
2	1	1	0
5	4	1	1
3	2	1	0

→ x_1	x_2	x_3	y	z	ω	s	y_p	y	Error
6	5	1	1	0.2	0.768	0.384	0.594	1	0.08
2	1	1	0	0.4	0.598	0.299	0.574	0	0.164
5	4	1	1	1.0	0.731	0.345	0.590	1	0.084
3	2	1	0	0.6	0.648	0.322	0.579	0	0.167

→ 0.082 → 0.167